When is quantum error mitigation worth paying for?

Q-TRACE: selective error mitigation for noisy quantum fraud classifiers

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M.S. in Artificial Intelligence capstone · Katz School of Science and Health, Yeshiva University
Progress presentation · October 13, 2026

Mitigation is expensive, so use it selectively



Readout mitigation

Inverts measurement errors. About two calibration circuits per batch: almost free.



Zero-noise extrapolation (ZNE)

Amplifies gate noise and extrapolates back: 3× shots and 9× two-qubit-gate exposure.



Extra shots

3× shots on the same circuit: lowers sampling variance, not bias.



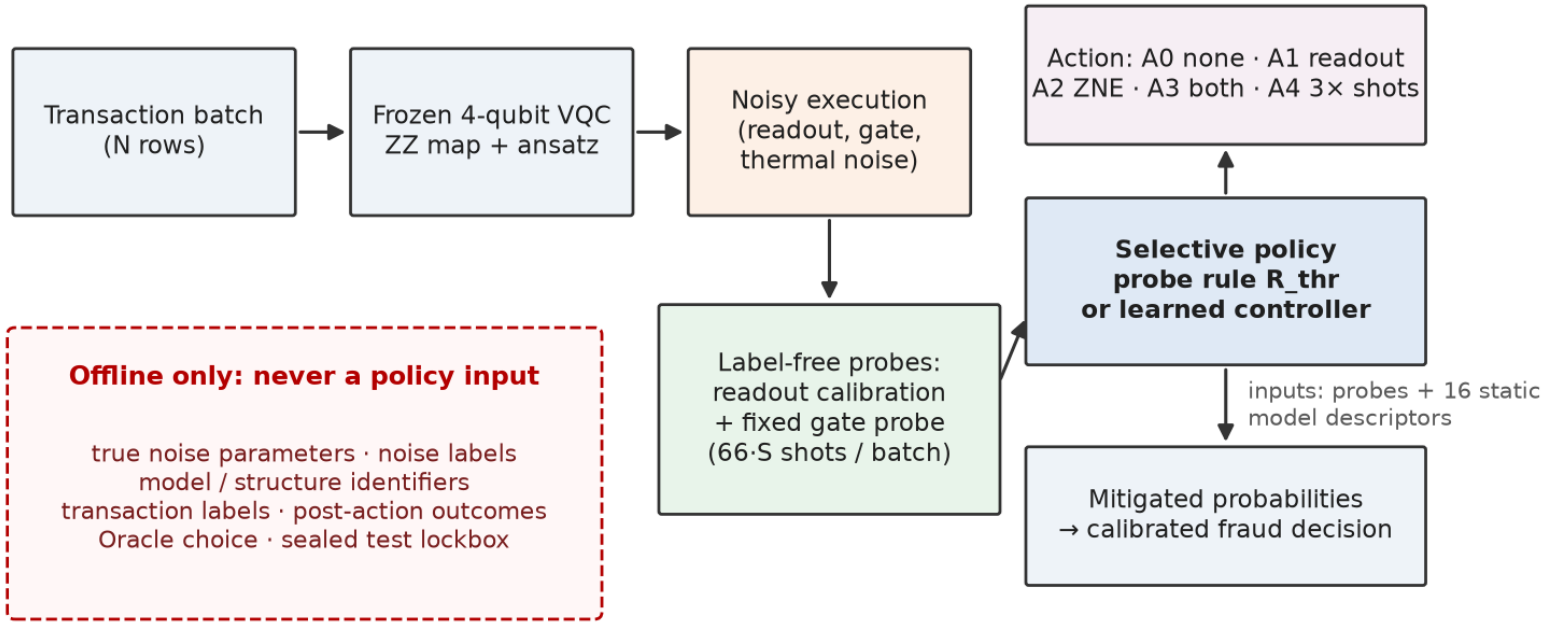
Original hypothesis

A learned controller (Q-TRACE), seeing only cheap label-free probes, chooses none / readout / ZNE / both / extra shots per context better than fixed policies.

Pre-registered alternative: a simple rule is enough.

Setup: 64 noisy quantum fraud classifiers

Q-TRACE decision layer: what the policy sees and what it may choose



283,726

credit-card transactions, 0.17% fraud, in 5 leakage-safe partitions

64

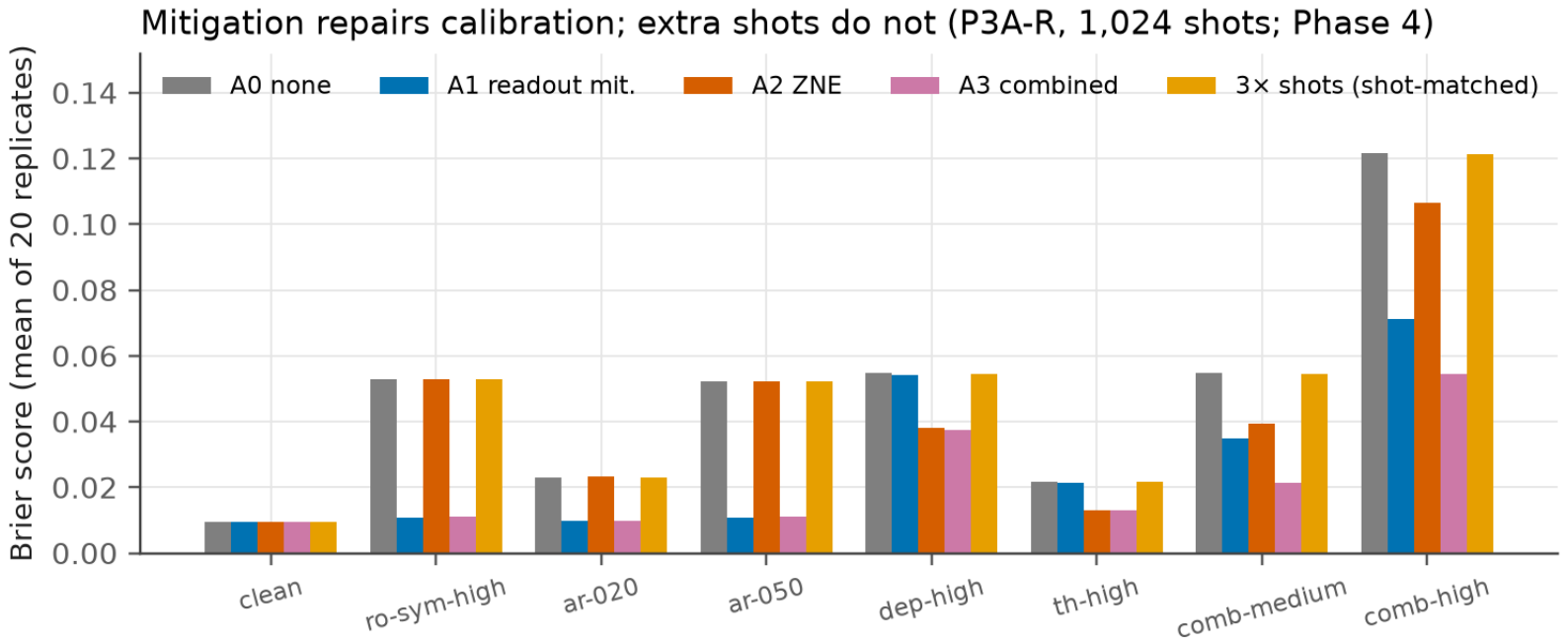
independent 4-qubit VQCs (16 structures × 4 seeds), frozen

1.4 M

counterfactual outcomes: every model × noise × budget × action

Policies see only label-free probes and static model descriptors; true noise, labels and the sealed test are offline-only.

Noise breaks calibration, and mitigation repairs it



Brier score by condition and action (1,024 shots, 20 replicates; Phase 4).

≤ 0.003

change in AUPRC under gate noise: discrimination survives

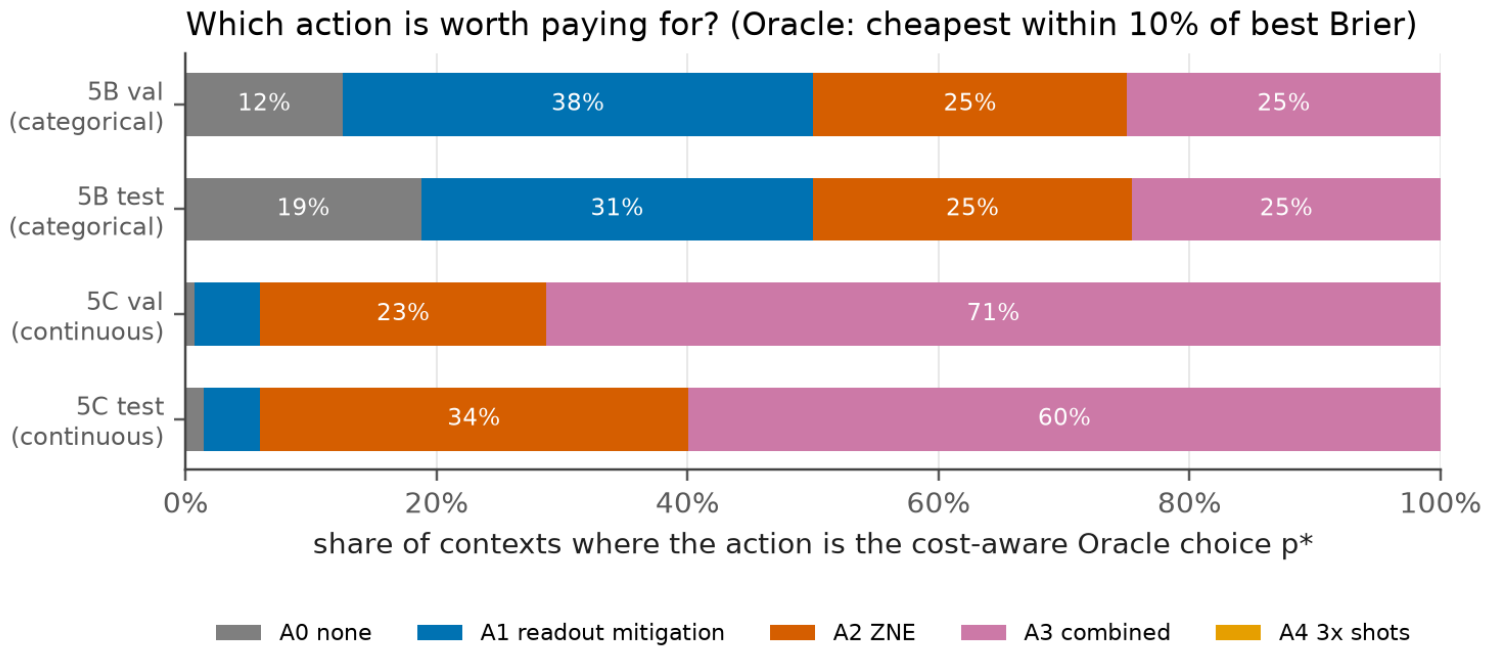
13×

worse Brier under combined noise (0.009 → 0.121)

0.122 → 0.054

Brier with combined mitigation; 3× shots leaves it at 0.121

Two pre-registered gates: nothing material left to learn



Share of contexts where each action is the cost-aware best choice (cheapest within 10% of the best Brier).

Phase 5B · categorical noise
Best action = lookup on noise type. A 4-cell probe rule is right in **99.7%** of validation cases.

Phase 5C · continuous joint noise
ZNE or combined mitigation wins in 94% of contexts. A two-threshold rule makes material errors in **2.0%** of cases.

Sealed held-out test: the rule matches the Oracle

2.0%

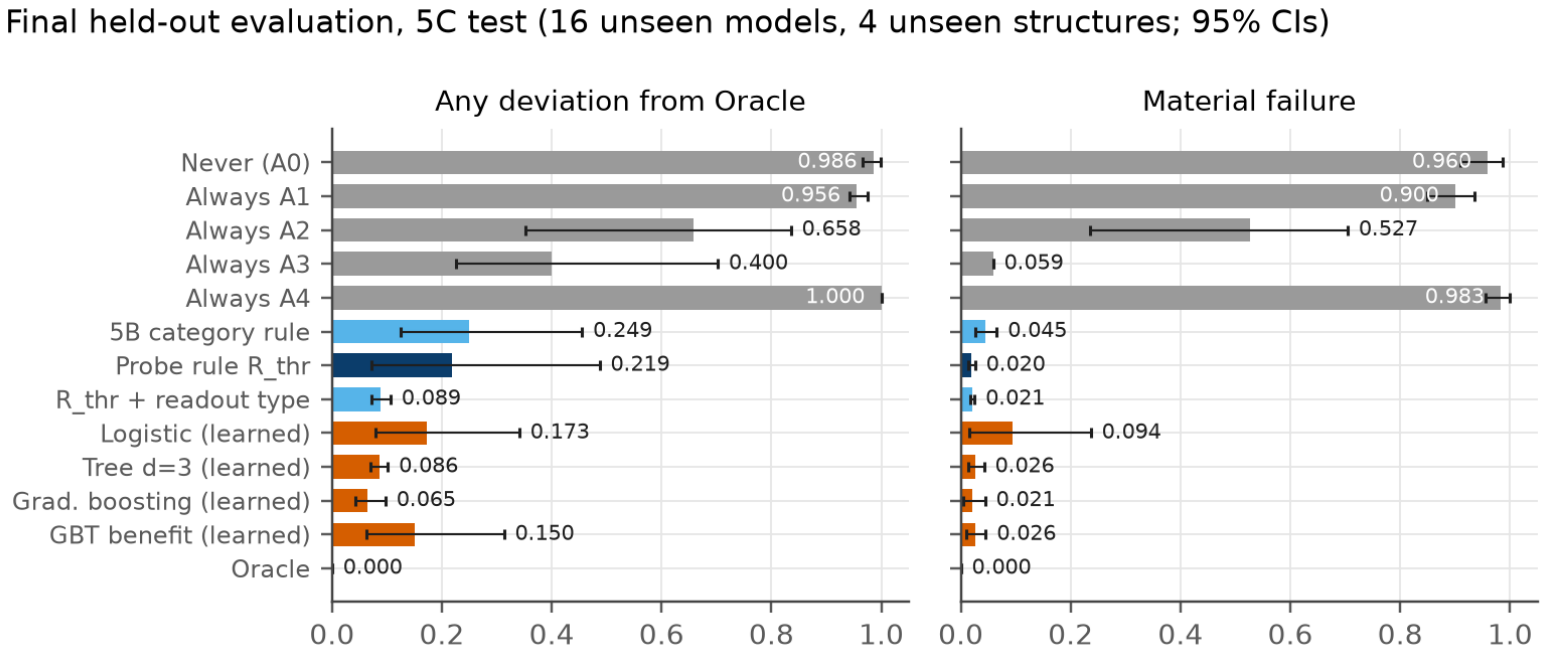
material errors on 16 unseen models (95% CI 1.5–2.6%)

−0.03%

Brier regret vs the cost-aware Oracle

+0.6%

shots vs the Oracle; 3.4% fewer than always-combined



One-time evaluation of frozen policies on sealed test structures (continuous joint noise). Best learned comparator: 2.1% material errors.

Why the simpler policy prevailed



Bias, not variance

Mitigation fixes a calibration bias that extra shots cannot remove, so the shot budget barely matters.



One question decides

In our design, 16 of 17 gate settings carry gate noise, so the material choice is whether to pay for ZNE. The gate probe alone recovers the rule.



Models differ in magnitude, not in the winner

Benefit varies a lot between models but rarely changes the winner. On an unseen readout type, learned models made 5–30% material errors; the rule 1.4%.

Contribution and remaining academic work

What we contribute

- Mitigation fixes calibration, not discrimination; extra shots never substitute.
- A leakage-safe benchmark: 64 VQCs × noise × budget × 5 actions.
- A pre-registered test: a simple probe rule is enough here; fixed learned comparators did not reduce material errors.

Planned milestones (to confirm with the course)

Oct 20	Polished research manuscript
Oct 27	Updated capstone presentation
Nov 10–24	Revisions from professor feedback
Dec 8	Final presentation package